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POLISHING PAD CONDITIONER POSITION DETECTION MODULE, POLISHING DEVICE, AND POLISHING DETECTION METHOD USING THE POSITION DETECTION MODULE

Abstract

A polishing device includes a rotatable platen; a polishing pad placed on an upper surface of the platen and having a polishing surface; a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head; a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad; a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad; a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0034061 filed in the Korean Intellectual Property Office on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

[0002] The present disclosure relates to a polishing pad conditioner position detection module, a polishing device, and a position detection method using the position detection module.

(b) Description of the Related Art

[0003] The chemical mechanical polishing or planarization CMP process of semiconductors is a process of flattening the wafer surface using chemical reactions and mechanical forces. It is a precise process in which not only mechanical factors such as the rotational speed of a polishing pad and a wafer, the pressure applied to the wafer, and the pattern directionality of the pad, but also chemical influences such as the interaction between slurry abrasive particles and a wafer surface and the role of the slurry organic additives are important variables.

[0004] On the point of polishing quality by the CMP process, the distribution of abrasive particles that are spread and maintained throughout the entire CMP polishing pad is important. In order to secure the performance of the CMP polishing pad, a process of conditioning the CMP polishing pad using a polishing pad conditioner is often required, and the conditioning process may include conditioning the upper part of the CMP polishing pad by placing the conditioner disk mounted on the polishing pad conditioner facing the CMP polishing pad.

[0005] However, in the process of conditioning the CMP polishing pad, there are instances where the conditioner disk mounted on the conditioner's holder may detach from the disk holder and also, during the movement of the conditioner arm equipped with a conditioner disk, issues such as degradation in the connecting components used to move the conditioner arm may occur, leading to errors in the movement position of the conditioner arm.

[0006] Accordingly, as the conditioner disk that is not placed in correct place on the CMP polishing pad conditions the CMP polishing pad there is a problem that uneven wearing occurs in the CMP polishing pad.

SUMMARY OF THE INVENTION

[0007] The present disclosure addresses some of these issues, and provides a polishing pad conditioner position detection module, a polishing device, and a position detection method using a position detection module, which can help prevent the deterioration of the polishing pad during the conditioning process by determining whether the disk is mounted on the holder of the conditioner arm and whether the conditioner arm is in correct position through a detector fixed to the cleaning module for cleaning the disk.

[0008] According to an example embodiment, a polishing device includes a polishing pad conditioner position detection module, a polishing pad conditioner, a disk for conditioning a polishing surface on a polishing pad, and a cleaning module for cleaning the disk and placed adjacent to the polishing pad. The polishing pad conditioner position detection module includes a frame configured to surround the cleaning module; a housing connected to the frame; and a detector disposed inside the housing and configured to face a side of the disk when the disk is attached to the polishing pad conditioner. The detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in a correct

predetermined position.
[0009] According to an embodiment, a polishing device includes a rotatable platen; a polishing pad placed on an upper surface of the platen and having a polishing surface; a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head; a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad; a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad; a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.
[0010] According to an embodiment, a polishing pad conditioner position detection method for detecting whether a polishing pad conditioner including a conditioner arm, a head connected to the conditioner arm, and a disk mounted on the head is in correct position, includes: placing the polishing pad conditioner in cup position adjacent to a cleaning module for cleaning the disk; and detecting, using a detector placed on the cleaning module, whether a center of the disk is located within a predetermined distance in a first direction from the detector.
[0011] According to embodiments, when the polishing pad conditioner is placed in a cup position close to the cleaning module that cleans the conditioner disk, the detection part connected to the cleaning module detects whether the disk is mounted on the conditioner and whether there is a problem with the movement of the conditioner arm, thereby improving the thickness quality of the polishing pad by preventing the wearing down of the polishing pad during the conditioning process.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.
[0013] FIG. **2** is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.
[0014] FIG. **3** is a diagram illustrating a cross-section of a polishing pad conditioner position detection module according to an embodiment.
[0015] FIG. **4** is a diagram illustrating an exploded state for describing each configuration of a polishing pad conditioner position detection module according to an embodiment.
[0016] FIG. **5** is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.
[0017] FIGS. **6**A-**6**C are diagrams for describing a polishing pad conditioner position detection module and method according to an embodiment.
[0018] FIG. **7** is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the disk is separated.
[0019] FIG. **8** is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the disk is separated.
[0020] FIG. **9** is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in correct position.
[0021] FIG. **10** is a diagram for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in a correct position.
[0022] FIG. **11** is a view for describing a polishing device according to an embodiment.
DETAILED DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so that those with ordinary skilled in the technical field to which the present disclosure belongs can easily implement it. The present disclosure may be implemented in various different forms and is not limited to the embodiments described herein.
[0024] In order to clearly describe the present disclosure in the drawings, certain parts which may be known in the art are omitted, and the same reference numerals are added to the same or similar components throughout the specification.
[0025] In addition, the size and thickness of each component shown in the drawing are arbitrarily expressed for convenience of explanation, so this disclosure is not necessarily limited to the illustration. In the drawings, for convenience of explanation, the thickness of layers, films, panels, regions, etc., are exaggerated for clarity.
[0026] Throughout the specification, it will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element (or using any form of the word “contact”), there are no intervening elements present at the point of contact. In addition, unless explicitly described to the contrary, the word “comprise”, and variations such as “comprises” or “comprising”, will be understood to imply the inclusion of stated elements but not the exclusion of any other elements. Furthermore, when a component is described as “including” a particular element or group of elements, it is to be understood that the component is formed of only the element or the group of elements, or the element or group of elements may be combined with additional elements to form the component, unless the context indicates otherwise. The term “consisting of,” on the other hand, indicates that a component is formed only of the element(s) listed.
[0027] In addition, when a part of a layer, film, region, plate, etc. is “above” or “on” another part, the reference part may be located above or below the reference part, depending on orientation, and does not necessarily mean “above” or “on” in the opposite direction of gravity.
[0028] In addition, throughout the specification, a plan view refers to a view from above, and a cross section view refers to a view from the side, unless the context indicates otherwise.
[0029] Ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first”) in a particular claim may be described elsewhere with a different ordinal number (e.g., “second”) in the specification or another claim.
[0030] Also these spatially relative terms such as “above” and “below” as used herein have their ordinary broad meanings—for example element A can be above element B even if when looking down on the two elements there is no overlap between them (just as something in the sky is generally above something on the ground, even if it is not directly above).
[0031] In the process of conditioning the CMP polishing pad of a CMP device, the conditioner disk mounted on the holder of the conditioner sometimes may deviate from the conditioner holder. Therefore, as the conditioning process proceeds on the CMP polishing pad without the conditioner disk being properly mounted on the holder, the quality of the CMP polishing pad may deteriorate.
[0032] In addition, in the process of moving the conditioner disk to face the CMP polishing pad while the conditioner arm is equipped with the conditioner disk, there are cases where the conditioner arm is moved to a position different from the target position due to problems such as deterioration of the parts related to the above-mentioned movement. In this case, during the conditioning process, the conditioner disk mounted on the conditioner arm is placed to be biased to one side on the CMP polishing pad, causing uneven abrasion of the CMP polishing pad.
[0033] The polishing pad conditioner position detection module **10** according to the present

disclosure is configured to detect whether the detection part **400** is equipped with the disk **130** in the conditioner **100** and whether there is a problem with the movement of the conditioner arm **140**.
[0034] Specifically, the detection part **400** is configured to determine whether the disk **130** is detached by checking whether the disk **130** is mounted on the polishing pad conditioner **100**. For example, while the polishing pad conditioner position detection module **10** is placed in the cup position (e.g., a position where it is above a stationary cup that collects residue from a cleaning process), by checking whether the conditioner arm **140** is placed in a correct position and whether the disk **130** is detected within a predetermined range, it may be possible to determine whether there is an error in the movement of the conditioner arm **140**.
[0035] According to some embodiments, the polishing pad conditioner position detection module **10** according to this disclosure first determines whether the disk **130** is mounted and positioned correctly before conditioning, and allows conditioning only if there are no issues, thereby preventing the polishing pad **1** from uneven abrasion during the conditioning process and thereby improving the thickness quality of the polishing pad **1**.
[0036] Hereinafter, with reference to the drawings, a polishing pad conditioner position detection module **10**, a polishing device **20**, and a detection method using the position detection module according to an embodiment of the present disclosure will be described in more detail.
[0037] FIG. **1** is a diagram illustrating a polishing pad conditioner position detection module according to an embodiment, in which a head **110** and a disk **130** of the polishing pad conditioner **100** are shown in cup position adjacent to a cleaning module **200**.
[0038] The cup position refers to the position where the disk **130** of the polishing pad conditioner **100** is placed when it is not being used to condition the top of the polishing pad **1** before conditioning the polishing surface **2**.
[0039] For example, the cup position refers to a position in which the head **110** of the polishing pad conditioner **100** is located adjacent to the cleaning module **200** positioned next to the polishing pad **1** and is maintained in a standby mode before conditioning. The cup position may be a set standby position or a non-polishing position. For example, the cup position may refer to a predetermined set position in which the disk **130** and head **110** do not overlap the polishing pad **1** in a plan view. When the disk **130** is in the cup position, the cleaning module **200** may clean the disk **130** of the polishing pad conditioner **100**.
[0040] As shown in FIG. **1**, the polishing device **20** in this disclosure may include a rotatable platen **22**, a polishing pad **1** disposed on the upper surface of the platen and having a polishing surface **2**, a polishing head **24** disposed on the top of the polishing pad and mounting a substrate **3** on the bottom, a driving part **26** that rotates the platen **22** with respect to an axis perpendicular to the polishing pad **1**, a slurry supplying part **28** (e.g., an outlet such as a nozzle connected to a slurry supply) that supplies slurry to the upper surface of the polishing pad **1**, and a polishing pad conditioner **100** that conditions the polishing surface **2** of the polishing pad **1**.
[0041] Referring to FIG. **1**, in a polishing device **20** which includes a polishing pad conditioner **100** with a disk **130** for conditioning the polishing surface **2** and a cleaning module **200** positioned adjacent to the polishing pad **1** for cleaning the disk **130**, the disclosed polishing pad conditioner position detection module **10** according to a present disclosure is configured to detect the position of the polishing pad conditioner **100** when the polishing pad conditioner **100** is in the cup position.
[0042] Specifically, the polishing pad conditioner position detection module **10** detects the position of the polishing pad conditioner **100** in the state where the polishing pad conditioner **100** is in the standby position. In some embodiments, the polishing pad conditioner **100** includes a holder **120** located under the head **110** of polishing pad conditioner **100**. A disk **130** may be mounted on the holder **120** for conditioning the surface of the polishing pad **1**. The disk **130** may be positioned near the cleaning module **200** for cleaning the disk **130** in a cup position.
[0043] As shown in FIG. **1**, the head **110** is provided at the other end of the conditioner arm **140**, and the disk **130** is mounted on the lower part of the head **110**.

[0044] FIG. **1** shows a state in which the disk **130** is mounted on the head **110**. Accordingly, the holder **120** disposed on the lower surface of the head **110** is omitted and does not appear in the drawing. The structure and arrangement position of the holder **120** can be seen in FIGS. **7** and **8**.
[0045] The conditioning process is such that the disk **130** mounted on the conditioner arm **140** (specifically, the disk **130** mounted on the head **110**) is placed to face the polishing pad **1**, and the disk **130** is polished. Conditioning the upper part of the polishing pad **1**.
[0046] First, the disk **130** presses the polishing pad **1** and rotates around the center of the disk **130** on an axis. Due to the pressure and rotation of the disk **130**, friction occurs between the surface where the disk **130** and the polishing pad **1** come into contact. By the above friction, the polishing pad **1** is conditioned.
[0047] The process of the disk **130** pressing the polishing pad **1** includes the disk **130** descending toward the polishing pad **1**, and the disk **130** moves until it contacts the polishing pad **1**. Afterwards, the pressure applied to the disk **130** may be adjusted through a separate control unit depending on the degree of conditioning.
[0048] The process by which the disk **130** descends is as follows.
[0049] According to one embodiment, while the conditioner arm **140** and the head **110** connected to the conditioner arm **140** move up and down as a whole, the disk **130** mounted on the head **110** may move up and down. At this time, the drive shaft **150** supporting the conditioner arm **140** may have a structure whose length can be adjusted in the vertical direction, and the height at which the conditioner arm **140** is placed can be adjusted according to the length adjustment of the drive shaft **150**.
[0050] As another example, the position (height) at which the conditioner arm **140** is disposed may be fixed. In this case, the head **110** connected to the conditioner arm **140** may have a structure capable of moving up and down.
[0051] Specifically, the head **110** is positioned in contact with the other end surface of the conditioner arm **140** (raised state, state in FIG. **1**) and a position spaced a certain distance from the other end surface of the conditioner arm **140** (down state). The head **110** can move up and down. Here, when the head **110** is lowered, the disk **130** mounted on the head **110** is placed in contact with the polishing pad **1**.
[0052] As another embodiment, the holder **120** provided on the lower surface of the head **110** may be moved up and down while maintaining the height of the conditioner arm **140** and the head **110**. The process of the disk **130** applying pressure to the polishing pad **1** is a general process and may be similar to the process performed in an existing conditioning device. Information related to this is disclosed in Korean Patent Publication No. 10-2020-0043209, etc.
[0053] According to the present disclosure, the disk **130** is placed in contact with the holder **120** provided below the head **110**. At this time, the holder **120** serves to couple the disk **130** to the lower part of the head **110**. Referring to FIGS. **7** and **8**, the holder **120** has a structure that protrudes from the lower part of the head **110**.
[0054] According to an embodiment, the holder **120** may have a structure integrated with the head **110**.
[0055] According to another embodiment, the length of the holder **120** protruding from the lower surface of the head **110** may be adjustable. At this time, the holder **120** may have a structure that allows it to move up and down. In this case, one side of the holder **120** may move up and down through the inside of the head **110**.
[0056] The protruding portion of the holder **120** comes into contact with the upper surface of the disk **130** and serves to hold the disk **130**. In other words, the holder **120** is coupled to the disk **130** and serves to couple the disk **130** to the head **110**.
[0057] Referring to FIG. **8**, the lower surface of the head **110** and the holder **120** have a step. Accordingly, the upper surface of the disk **130** may have a structure with a step so as to be in close contact with the step.

[0058] As shown in FIG. **1**, the disk **130** may have a plate-shaped structure with a thickness. The disk **130** may be based on a metal base material and may have a structure including a polishing portion (a surface in contact with the polishing pad **1**) where a plurality of diamond abrasive are adhered to the surface of the metal base material.
[0059] The disk **130** according to the present disclosure may have a structure including the step described above so that the opposite side of the polishing portion of the disk **130** is mounted on the holder **120** and the head **110**.
[0060] The method of coupling the disk **130** and the holder **120** may be a magnetic attachment method using magnetic force, or a method using selective engagement of grooves and protrusions by rotation, etc. may be applied. The disk **130** may be coupled to the holder **120** in various ways and thus may be mounted on the head **110** provided with the holder **120**.
[0061] Referring to FIGS. **7** and **8**, the holder **120** may have a cylindrical shape. However, the structure of the holder **120** is not limited to the structure shown. The structure of the holder **120** is not limited as long as it is disposed on the lower surface of the head **110** and coupled to the upper surface of the disk **130**.
[0062] As shown in FIG. **1**, the head **110** may have a cylindrical shape. However, the shape of the head **110** is not limited to the shape shown in FIG. **1**. The structure of the head **110** may vary depending on the structure of the holder **20** and the disk **130**.
[0063] Although not shown in the drawing, the polishing pad conditioner **100** may further include a head driving unit that rotates the head **110**. The head driving unit may rotate the head **110** based on the axis of the head **110**.
[0064] Referring to FIG. **1**, the head **110** can rotate based on the head **110** axis, and the holder **120** provided on the lower surface of the head **110** also rotates with the rotation of the head **110**. At this time, the disk **130** coupled to the holder **120** and mounted on the lower part of the head **110** also rotates around the center of the disk **130**.
[0065] In FIG. **1**, as mentioned above, when the polishing pad conditioner position detection module **10** is placed in the cup position, the polishing pad **1** is not disposed below the disk **130** to overlap the disk **130** in the plan view.
[0066] In the cup position as shown in FIG. **1**, the polishing pad conditioner position detection module **10** checks whether the disk **130** mounted on the polishing pad conditioner **100** is detected in correct position. Then to carry out the conditioning process, the polishing pad conditioner **100** is moved so that the disk **130** mounted on the conditioner arm **140** is positioned over the polishing surface **2** of the polishing pad **1** as the conditioner arm **140** rotates around the drive shaft **150** (refer to FIG. **11**).
[0067] As shown in FIG. **1**, a polishing pad conditioner **100** may include a drive shaft **150** disposed in a direction perpendicular to a bottom surface of disk **130**, a conditioner arm **140** having one end (a first end) connected to the drive shaft **150**, a head **110** connected below the other end (a second end opposite the first end) of the conditioner arm **140**, a holder **120** disposed on the lower surface of the head **110**, and a disk **130** mounted on the holder **120** to condition the polishing surface **2** that is the upper surface of the polishing pad **1**.
[0068] The polishing pad conditioner position detection module **10** according to embodiments of the present disclosure is connected to the cleaning module **200** and maybe positioned under the injection nozzle **210** of the cleaning module **200** as shown in FIG. **1**. Here, the injection nozzle **210** of the cleaning module **200** serves to clean the disk **130** of the polishing pad conditioner **100**, and the disk **130** may be cleaned by spraying liquid such as DIW (De-ionized Water) toward the disk **130** through the injection nozzle **210**.
[0069] The injection nozzle **210** is disposed on one side of the cleaning module **200** facing the disk **130**. In some embodiments, the cleaning module **200** may include an injection nozzle **210** for spraying DIW towards a side of the disk **130** and an injection part (which can be a second nozzle, not shown) disposed to spray DIW onto a lower surface of the disk **130** with the spray direction

facing the lower surface of the disk **130**. The injection part also serves to clean the disk **130**.

[0070] Here, the side of the disk **130** means the outer periphery of the disk **130**, which may include an outer sidewall of the disk **130**, the portion mounted on the holder **120** is defined as the upper surface of the disk **130**, and the opposite surface facing the upper surface is defined as the lower surface of the disk **130**.

[0071] The polishing pad conditioner position detection module **10** according to some embodiments of the present disclosure may include a housing **300** placed on the outer surface of the cleaning module **200** and a detection part **400** placed inside the housing **300** so that one end faces the side of the disk **130**, and by checking whether the disk **130** is detected within a predetermined distance through the detection part **400**, it is possible to check whether the disk **130** is mounted on the holder **120**.

[0072] FIG. **1** shows an embodiment of the polishing pad conditioner position detection module **10**, and the polishing pad conditioner position detection module **10** may include a housing **300** disposed to be connected to one surface (e.g., one side) of the cleaning module **200** and a detection part **400** disposed inside the housing **300**, and the housing **300** may include a protruding portion in which the detection part **400** is disposed.

[0073] In FIGS. **1-5**, the housing **300** is connected to one side of the cleaning module **200** and is placed to protrude from the cleaning module **200** in the direction in which the disk **130** is located.

[0074] During detection, a first end of the detection part **400** disposed inside the housing **300** faces the side of the disk **130**, and the first end of the detection part **400** detects the disk **130**, so that it is possible to check whether the disk **130** is properly mounted in the holder **120**.

[0075] Although FIGS. **1-5** show a structure in which the housing **300** is connected to one surface of the cleaning module **200**, a particular manner in which the housing **300** of the polishing pad conditioner position detection module **10** according to the present disclosure is fixed to the cleaning module **200** is not limited to this example.

[0076] In the drawings and embodiments to be described below, the housing **300** is placed on at least one side of the cleaning module **200** and further includes a supporting part **330** for fixing the housing **300** to the cleaning module **200**. The supporting part **330** may surround all four side surfaces of the cleaning module **200**.

[0077] The cable **500** connected to the detection part **400**, which will be described below, is disposed inside the housing **300** and may have a structure electrically connected to the outside of the polishing pad conditioner position detection module **10** and cleaning module **200**.

[0078] FIG. **2** is a diagram for describing a polishing pad conditioner position detection module according to an embodiment, FIG. **3** is a diagram illustrating a cross-section of a polishing pad conditioner position detection module according to an embodiment, and FIG. **4** is a diagram illustrating an exploded state for describing each configuration of a polishing pad conditioner position detection module according to an embodiment. FIG. **5** is a diagram for explaining a state in which the polishing pad conditioner position detection module **10** is coupled to the cleaning module **200**.

[0079] First, FIG. **2** is a diagram showing an example perspective view of the polishing pad conditioner position detection module **10**, and FIG. **3** is a diagram showing a cross-section seen from above of the polishing pad conditioner position detection module **10**. Through FIG. **3**, an example arrangement structure of the detection part **400** and the cable **500** disposed inside the polishing pad conditioner position detection module **10** are depicted.

[0080] FIG. **4** is an exploded view of the polishing pad conditioner position detection module **10**, and it is possible to confirm the structure in which each component of the polishing pad conditioner position detection module **10** is combined.

[0081] As shown in FIG. **2** to FIG. **4**, the polishing pad conditioner position detection module **10** according to the present disclosure includes a housing **300** placed on the outer surface of the cleaning module **200**, a detection part **400** placed inside the housing **300** so that one end faces the side of the disk **130**, and a supporting part **330** that fixes the housing **300** to the cleaning module **200**, and the detection part **400** detects whether the disk **130** is mounted in the holder **120** of the polishing pad conditioner **100**.

[0082] The supporting part **330** may be a support frame that is disposed to surround the periphery of the cleaning module **200** and that supports the housing. At least one surface of the supporting part **330**, such as an outer surface of a side beam, is coupled to the housing **300** to fix the housing **300** to the cleaning module **200**. The supporting part **330** may be a frame including four sides, each side being formed by a beam (e.g., a side beam). As depicted in FIGS. **2-4**, some of the beams may be integrally formed and/or permanently affixed to each other, and at least one beam may be detachably connected to the other beams, for example using screws or a nut and bolt system or other attachment components. The beams may form a rectangular shape having inner side surfaces and outer side surfaces. The beams that form the supporting part **330** may have a rigid structure and may be formed of a hard, rigid material such as a hard plastic.

[0083] Due to the rigid structure and the surrounding frame, the polishing pad conditioner position detection module **10** fixed by the supporting part **330** may be constructed so that it does not shake or so it minimizes any shaking even with external vibration, so that when the detection part **400** detects the position of the disk **130** and the conditioner arm **140**, an error caused by the shaking of the polishing pad conditioner position detection module **10** may be minimized.

[0084] In some embodiments, the housing **300** may be formed of two joined bars or plates that have hollowed-out sections between them which house certain components. For example, inside the housing **300**, a detection part **400** and a cable **500** connected to the detection part **400** may be disposed.

[0085] The detection part **400**, which may be a detector such a sensor, may be disposed inside a compartment formed in the housing **300** so that when the polishing pad conditioner position detection module **10** is positioned to be adjacent to the head **110** and disk **130**, one end (e.g., a first end) faces the side surface of the disk **130**, and the cable **500** may be connected to another end (e.g., a second, opposite end) of the detection part **400** to be disposed inside the housing **300**.

[0086] The cable **500** has one end (e.g., a first end) connected to the second end of the detection part **400** and another (e.g., a second, opposite end) end extending to the outside of the housing **300**, and serves to electrically connect the detection part **400** to the outside of the housing **300**.

[0087] The cable **500** may include a first cable portion **510** connected to the second end of the detection part **400** and placed inside the housing **300**, and a second cable portion **520** connected to the first cable portion **510** and extending outside the housing **300**.

[0088] The polishing pad conditioner position detection module **10** according to the present disclosure may further include a receiving part **350** connected to the housing **300** to accommodate the cable **500**. The receiving part **350** may further include a hole **600** for routing the second cable portion **520** externally.

[0089] As shown in FIGS. **2** to **4**, one surface of the receiving part **350** (e.g., a lower outer surface facing the supporting part **330**) may be coupled to the supporting part **330**, and another surface of the receiving part **350** (e.g., a bottom surface of a bottom panel of the receiving part **350**) may separately be coupled to the housing **300**. Though not shown, the receiving part **350**, which may be a compartment that houses the cable, may be connected to the supporting part **330** using attachment components such as screws or a nut and bold system, and may be connected to the housing **300** using attachment components such as screws or a nut and bold system. The housing **300**, and the receiving part **350** may also be fixed to the cleaning module **200**, as can be seen, for example, in FIG. **5**, which depicts the supporting part **330** surrounding the cleaning module **200**. The supporting part **330** (e.g., frame) may be affixed to the cleaning module **200** by surrounding outer surfaces of the cleaning module **200** and being tightened in place to securely attach to the cleaning module **200**. In one embodiment, connector **250**, which may be a connecting beam, connecting rod, or other rigid bar or arm-type structure, may connect the cleaning module **200** to the drive shaft

150, etc. For example, part of the drive shaft **150** may be stationary and non-rotating (or may be rotatable separately from a portion that moves the conditioner arm **140**), and may be affixed to a beam, rod, bar, or arm that connects to the cleaning module **200**.

[0090] The detection part **400** and the cable **500** may be protected by being disposed inside the housing **300** and the receiving part **350**.

[0091] An injection nozzle **210** may be attached on an outer surface of the cleaning module **200** and may protrude beyond an outer surface of the cleaning module **200**. For example, the cleaning module **200** may have a parallelepiped shape, for example, in the form of a box having six sides, and may house cleaning pipes that transport cleaning solution to the injection nozzle **210**. The cleaning solution may be received from outside the cleaning module **200** through piping, or may be included in the cleaning module **200**, and spraying of the cleaning solution through the nozzle **210** may be controlled by a controller. The cleaning module **200** may be a spray dispenser. The detection part **400** may have a structure that protrudes from the cleaning module **200** more than the injection nozzle **210** in the same direction as the direction in which the injection nozzle **210** of the cleaning module **200** protrudes. The greater protruding amount of the protruding detection part **400** may result in the protruding detecting part **400** being disposed closer to the disk **130** when the disk **130** is disposed adjacent to the cleaning module **200**.

[0092] In this case, the housing **300** in which the detection part **400** is disposed also has a protruding structure together with the detection part **400**, and the detection part **400** is disposed inside the housing **300**. Therefore, the detection part **400** and the housing **300** in which the detection part **400** is disposed may be placed to protrude from the cleaning module **200** toward the disk **130**.

[0093] This arrangement may prevent the DIW from also sticking to the detection part **400** located below the injection nozzle **210** in the process of spraying the DIW to clean the disk **130** in the injection nozzle **210**. For example, by having a structure in which the housing **300** surrounds the protruding detection part **400**, it is possible to protect the detection part **400** from DIW.

[0094] As shown in FIG. **4**, the housing **300** may include a first housing portion **310** in which a detection part **400** and a cable **500** are disposed on its upper surface and a second housing portion **320** coupled to the top of the first housing **310**. According to an embodiment, the first housing portion **310** and the second housing portion **320** may be coupled using screws, or nuts and bolts. Each of the first housing portion **310** and second housing portion **320** may be a plate, beam, or bar. Though shown to have a flat top surface, second housing portion **320** is not limited to this structure and shape and can have other structures or shapes. Similarly, though shown to have a flat bottom surface, first housing portion **310** is not limited to this structure and shape and can have other structures or shapes. The first housing portion **310** may be a first casing, or lower casing. The second housing portion **320** may be a second casing, or upper casing.

[0095] The first housing portion **310** may further include a guide part **340** for fixing the detection part **400** disposed therein.

[0096] Depending on the embodiment, the guide part **340** may also fix the first cable portion **510** connected to the detection part **400**. The guide part **340** may have a groove structure that follows the shape of the detection part **400** and the first cable portion **510**. The second housing portion **320** may also include a guide part **340**. When the first housing portion **310** and second housing portion **320** are connected together, the guide parts **340** together may form a compartment with the housing **300** through which the detection part **400** and first cable portion **510** pass.

[0097] FIG. **5** is a diagram for describing a polishing pad conditioner position detection module according to an embodiment.

[0098] FIG. **5** is a diagram for explaining a state in which the polishing pad conditioner position detection module **10** is coupled to the cleaning module **200**, and the second housing portion **320** is coupled the top of the first housing portion **310** in a state in which the first housing portion **310** and the supporting part **330** are coupled to the cleaning module **200**.

[0099] In one embodiment, a lower portion of the receiving part **350** (e.g., a lower case portion) is fixed to the cleaning module **200** in a state of being coupled to the housing **300** and the supporting part **330**, and an upper portion of the receiving part **350** (e.g., an upper case portion) may be coupled to overlap with the lower portion and fixed together.
[0100] Also, as shown in FIG. **5**, the detection part **400** may be disposed close to the disk **130**. The reason why the detection part **400** is disposed close to the disk **130** is to detect whether the target object, i.e., the disk **130**, is located within a predetermined distance. The detection part **400** may be disposed close to the disk **130** based on, for example, the conditioner arm **140** being rotated to be adjacent to the detection part **400**.
[0101] According to various embodiments, during detection by the detection part **400**, no other object is positioned between the detection part **400** and the disk **130**. Assuming that another object is located between the detection part **400** and the disk **130**, the detection part **400** may incorrectly determine the object detected by the detection part **400** as the disk **130**.
[0102] The detection part **400** may be a position detection sensor and may detect whether the disk **130** is located within a preset distance. For example, the detection part **400** may be a detector such as laser sensor, an image sensor, or another type of sensor that can sense or determine depth and/or distance.
[0103] When one end of the detection part **400** is arranged to face the direction in which the center of the disk **130** is located, if the direction is set to the first direction D**1**, the detection part **400** may detect whether the disk **130** is located within a predetermined distance in the first direction D**1**.
[0104] According to an embodiment, the detection part **400** may detect whether the disk **130** is located within a particular distance, such as 5 mm, from one end of the protruding detection part **400**.
[0105] In the state where the polishing pad conditioner **100** which includes a holder **120** disposed on the lower surface of the head **110** and a disk **130** mounted on the holder **120** to condition the upper surface of the polishing pad **1** is positioned in a cup position close to the cleaning module **200** that cleans the disk **130**, the polishing pad conditioner position detection module **10** according to the present disclosure may detect the position of part of the polishing pad conditioner **100**. The polishing pad conditioner position detection module **10** may include a housing **300** disposed on an outer surface of the cleaning module **200** and a detection part **400** disposed inside the housing **300** and oriented toward a first direction D**1** in which the center of the disk **130** is located.
[0106] The detection part **400** may detect whether a disk **130** is located within a predetermined distance in the first direction D**1**, and at the same time, whether a disk **130** is located within a predetermined width in the second direction D**2** perpendicular to the first direction D**1**.
[0107] With regard to determining the predetermined width in the second direction D**2**, the detection part **400** may determine whether the disk **130** is located in a predetermined position having a particular width. When the disk **130** is not detected in that position to cover the particular width, the disk **130** may be detected as being misaligned.
[0108] Referring to FIG. **9**, which will be described below, a detailed numerical description will be given. In FIG. **9**, large arrows and small arrows are shown in the D**2** direction, respectively. Please note that the large arrow is drawn to indicate that the distance is 5 mm in a straight line in the D**1** direction. Note that the small arrow is drawn to indicate a gap of 3 mm in the D**2** direction.
[0109] In the case of FIG. **9**, the disk **130** is detected within a certain distance (e.g., 5 mm) in the D**1** direction.
[0110] It is assumed that the shortest distance (straight line distance in the D**1** direction) from the sensing unit **400** to the disk **130** is 5 mm. In FIG. **9**, a large arrow in the D**2** direction is shown so as to contact the circumference of the disk **130**. The maximum protruding surface of the sensing unit **400** and the large arrow in the D**2** direction are arranged horizontally, and the vertical distance across the two horizontal straight lines is 5 mm.
[0111] The case of FIG. **9** corresponds to a case where the disk **130** is detected over a certain width

in the D**2** direction.

[0112] According to the above example, the width of the disk **130** (width in the D**2** direction) is detected within the 5 mm distance in the D**1** direction described above. Therefore, the disk **130** does not need to be detected at a position deviating from a distance of 5 mm in the D**1** direction.

[0113] The constant width in the D**2** direction, as shown in FIG. **9**, means the interval (width) indicated by the smaller arrow among the arrows shown in the D**2** direction. Here, according to one example embodiment, it is assumed that the distance pointed by the small arrow shown in the D**2** direction is 3 mm.

[0114] When referring to the above description, the disk **130** shown in FIG. **9** is detected within 5 mm in the D**1** direction, and when looking at the disk **130** at the exact position spaced by 5 mm in the D**1** direction, (that is, when looking at the disk **130** at the position where the large arrow in the D**2** direction is shown), at that position, the disk **130** is detected in a 3 mm width range in the D**2** direction.

[0115] For example, the polishing device **20** may be configured such that when the disk **130** is properly installed, and when the conditioner arm **140** is in a particular position, then the distance between the detection part **400** and the disk **130** in the first direction D**1** will be a known predetermined amount, and the disk will also be detected to have a known predetermined width in the second direction D**2**. Therefore, when the detection part **400** performs detection and the distance in the first direction D**1** and/or the width in the second direction D**2** does not match these predetermined amounts, then a misalignment may be detected, and either the disk **130**, the conditioner arm **140**, or both may be out of proper alignment.

[0116] The movement of the conditioner arm **140** as well as the operation of the detection part **400**, spraying of the cleaning liquid, and processing and comparison of various data to determine whether the disk **130** and conditioner arm **140** are properly positioned may be controlled by a controller. For example, the controller can include one or more of the following components: at least one central processing unit (CPU) configured to execute computer program instructions to perform various processes and methods, random access memory (RAM) and read only memory (ROM) configured to access and store data and information and computer program instructions, input/output (I/O) devices configured to provide input and/or output to the processing controller **1020** (e.g., keyboard, mouse, display, speakers, printers, modems, network cards, etc.), and storage media or other suitable type of memory (e.g., such as, for example, RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, flash drives, and any type of tangible and non-transitory storage medium) where data and/or instructions can be stored. In addition, the controller can include antennas, network interfaces that provide wireless and/or wire line digital and/or analog interface to one or more networks over one or more network connections (not shown), a power source that provides an appropriate alternating current (AC) or direct current (DC) to power one or more components of the controller, and a bus that allows communication among the various disclosed components of the controller.

[0117] FIG. **11** is a view for describing a polishing apparatus according to an embodiment.

[0118] As shown in FIG. **11**, the polishing apparatus **20** in this example may include a rotatable platen **22**, a polishing pad **1** disposed on the upper surface of the platen **22** and having a polishing surface **2**, a polishing head **24** disposed on the top of the polishing pad **1** and mounting the substrate **3** on the bottom, a driving part **26** that rotates the platen **22** with respect to an axis perpendicular to the polishing pad **1**, a slurry supplying part **28** that supplies slurry to the upper surface of the polishing pad **1**, and a polishing pad conditioner **100** that conditions the polishing surface **2** of the polishing pad **1**.

[0119] During a polishing operation, as depicted in FIG. **11**, as the platen **22** with the polishing pad **1** disposed on the top rotates about its axis, the polishing pad **1** also rotates together, and the polishing head **24** also rotates about the axis shown in FIG. **11**, so that the substrate **3** mounted below the polishing head **24** rotates at the same time as the platen.

[0120] As the substrate **3** in contact with the polishing surface **2** of the polishing pad **1** rotates at the same time as the polishing pad **1**, the substrate **3** in contact with the polishing surface **2** is polished, and slurry may be supplied from the slurry supplying part **28** during the polishing process.

[0121] As described in FIGS. **1** to **10**, the polishing pad conditioner **100** included in the polishing device **20** may include a conditioner arm **140** with one end connected to a drive shaft **150**. The conditioner arm may extend in a direction that is perpendicular to an axis around which the disk **130** is disposed. A head **110** may be connected below another end of the conditioner arm **140**, a holder **120** may be placed on the lower surface of the head **110**, and a disk **130** may be mounted on the holder **120** to condition the polishing surface **2** which is the upper surface of the polishing pad **1**. The disk **130** may be rotated and operated at the same time as the polishing head **24** rotating, so that the polishing pad **1** can be conditioned while polishing occurs.

[0122] The polishing pad conditioner position detection method using the polishing pad conditioner position detection module **10** according to the present disclosure is a method of detecting whether the polishing pad conditioner **100** is in correct position using the polishing pad conditioner position detection module **10** while the polishing pad conditioner **100** is placed in a particular position adjacent to the cleaning module **200** that cleans the disk **130**.

[0123] By performing the detection, it can be determined whether the disk **130** is misaligned or is separated from the rest of the polishing pad conditioner **100** and/or whether there is an error in the movement of the conditioner arm **140**.

[0124] Here, if the conditioner arm **140** is misaligned, for example, due to deterioration of the internal bearings disposed between the conditioner arm **140** and the driving shaft **150**, this may mean that the degree to which the conditioner arm **140** is rotated by the rotation of the driving shaft **150** is not as much as the targeted position or is rotated beyond the targeted position.

[0125] A polishing pad conditioner position detection method using the position detection module **10** according to the present disclosure, as described above, first determines whether the disk **130** is mounted in the holder **120** and positioned in correct position prior to conditioning. If there is no problem (for example, if the disk **130** is mounted in the holder **120** and there is no error in the movement of the conditioner arm **140**), by allowing the polishing pad conditioner **100** to condition the polishing pad **1**, the thickness quality of the polishing pad **1** is improved by preventing uneven abrasion of the polishing pad **1** during the conditioning process.

[0126] The polishing pad conditioner **100** may have a structure including a conditioner arm **140** with one end connected to a driving shaft **150** and a second end connected to the disk **130**, a head **110** connected below and affixed to the second end of the conditioner arm **140**, a holder **120** placed on the lower surface of the head **110**, and the disk **130** mounted on the holder **120** to condition the upper surface of the polishing pad **1**. For information on the structure and coupling method of the head **110**, the holder **120**, and the disk **130** mounted thereon, refer to the above description.

[0127] The polishing pad conditioner position detection module **10** according to embodiments of the present disclosure is a structure fixed to the cleaning module **200**, and when the polishing pad conditioner **100** is placed in a position close to the cleaning module **200**, it is also located close to the polishing pad conditioner position detection module **10**.

[0128] FIG. **6**A is a diagram depicting various electronic and physical components of a polishing pad conditioner position detection module **10**. FIGS. **6**B, **6**C, and **7**-**9** are diagrams for describing a method of detecting a position of the polishing pad conditioner **100**.

[0129] For example, FIG. **7** and FIG. **8** are diagrams for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the disk is separated.

[0130] FIG. **6**A shows components of the polishing pad conditioner position detection module **10**, such as for example, housing **300** (described previously), detection part **400** (described previously), decision part **700**, which may include a computer software and/or hardware, which receives input information (e.g., distance, width, image information, etc.) relating to the sensed data received from the detection part **400**, alarm part **800**, which may include a computer software and/or hardware, and may additionally include visual and/or auditory generating components (e.g., a light or speaker) that issues an alarm if the detection pert **400** detects an improperly positioned disk **130** (or a missing disk **130**), and a control part **900**, which may be or include the controller mentioned previously.

[0131] FIG. **6**B illustrates a method of detecting a position of a polishing pad conditioner **100**. In step S**10**, the polishing pad conditioner **100** is placed in a standby position adjacent to the cleaning module **200** for cleaning the disk **130**. Next, in step S**100**, it is determined whether the disk **130** is positioned within a predetermined distance along a first direction D**1** of the detection part **400** disposed on an outer surface of the cleaning module **200**.

[0132] For example, FIG. **7** illustrates a state in which the disk **130** is mounted on the polishing pad conditioner **100** positioned at the standby position. In addition, FIG. **7** shows a state in which the conditioner arm **140** is in a correct position and one end of the detection part **400** of the polishing pad conditioner position detection module **10** is placed adjacent to and facing the disk **130**.

[0133] However, if the conditioner arm **140** is not in the correct position and is tilted or otherwise misaligned, the disk **130** may not be located within a predetermined distance and the detection part **400** may detect a distance different from the predetermined distance.

[0134] Therefore, it is important to detect whether the conditioner arm **140** is in a normal position while the disk **130** is mounted on the polishing pad conditioner **100**.

[0135] As shown in FIG. **7**, the detection part **400** may be disposed to protrude from the cleaning module **200** toward the disk **130**, and one end of the detection part **400** may be positioned adjacent to the disk **130**. Within a predetermined distance indicated in FIG. **7**, the detection part **400** may sense whether the disk **130** is properly connected.

[0136] FIG. **7** is a case where one end of the detection part **400** placed inside the housing **300** is placed toward the first direction D**1** where the center of the disk **130** is located, and the disk **130** is located within a predetermined distance in the first direction D**1**. In this case, the detection part **400** may sense that the disk **130** is mounted on the holder **120** of the polishing pad conditioner **100** by the disk **130**.

[0137] As shown in step S**110** of FIG. **6**B in connection with FIG. **6**A, because the detected distance matches or is within a range of the predetermined distance in the first direction D**1**, decision part **700** determines that the holder **120** is equipped with a disk **130**. It should be noted that decision part **700**, alarm part **800**, and control part **900** may all be portions of a computer program that controls performance of all of the steps discussed herein. Therefore, though described separately to distinguish the different steps, these parts may be separate software modules, or may be combined into a single program.

[0138] On the other hand, FIG. **8** shows a case where one end of the detection part **400** disposed inside the housing **300** is arranged to face the first direction D**1** where the center of the disk **130** is supposed to be located, but the disk **130** may not be located within a predetermined distance in the first direction D**1**. For example, if the disk **130** is detached from the holder **120** of the polishing pad conditioner **100**, the disk **130** may not be detected by the detection part **400** since there may be no detected object within a predetermined distance, so it is determined that the disk **130** is detached from the holder **120** of the polishing pad conditioner **100**.

[0139] Therefore, as shown in FIGS. **6**B and **8**, when the disk **130** is not detected within a predetermined distance in the first direction D**1**, a step S**120** in which the judgment part **700** determines that the disk **130** is detached from the holder **120** or is improperly attached to the holder occurs.

[0140] If the disk **130** is not detected within a predetermined distance in the first direction D**1**, in step S**130** the alarm part **800** may generate an alarm.

[0141] In the method of detecting the position of the polishing pad conditioner **100** according to the present disclosure, in step S**200**, the detection part **400** detects whether the disk **130** is located within a predetermined distance in the first direction D**1** and within a predetermined width in the second direction D**2** perpendicular to the first direction D**1**.
[0142] The detection part **400** may detect whether the disk **130** is located within a predetermined distance in the first direction D**1** from the detection part **400** and whether the disk **130** is located within a predetermined location and has a predetermined width in the second direction D**2**.
[0143] In the case where the disk **130** is detected to be located within the predetermined location having the predetermined width in the second direction D**2**, and the disk **130** is determined to be properly aligned. If the disk **130** is detected only in a partial area within the predetermined width, the disk **130** is determined to be improperly aligned.
[0144] FIG. **6**C illustrates a method of detecting a position of a polishing pad conditioner **100**. In step S**10**, the polishing pad conditioner **100** is placed in a standby position adjacent to the cleaning module **200** for cleaning the disk **130**. Next, in step S**200**, it is determined whether the disk **130** is positioned within a predetermined distance in the first direction D**1** and within a predetermined width in a second direction D**2** perpendicular to the first direction.
[0145] As shown in FIG. **60**, when the disk **130** is detected within a predetermined distance in the first direction D**1** and within a predetermined width in a second direction D**2** perpendicular to the first direction, it is determined that the disk **30** and the conditioner arm **140** are placed in a proper position (S**210**). In this case, the polishing pad conditioner **100** may perform conditioning (S**220**).
[0146] Conversely, if the disk **130** is not detected within a predetermined distance in the first direction D**1** or the disk **130** is not detected within a predetermined width in a second direction D**2**, the disk **130** or the conditioner arm **140** is not placed at a proper position (S**230**). In this case, the alarm part **800** may generate an alarm (S**240**).
[0147] FIG. **9** and FIG. **10** are diagrams for describing a process in which a polishing pad conditioner position detection module according to an embodiment detects whether the conditioner is in a correct position.
[0148] FIG. **9** shows the case where the conditioner arm **140** is placed in correct position, and the polishing pad conditioner position detection module **10** may detect the disk **130** within a predetermined distance in the first direction D**1** and within a predetermined width in the second direction D**2** perpendicular to the first direction D**1**. That is, the disk **130** is detected within a predetermined distance in the first direction D**1** and within a predetermined width in the second direction D**2**.
[0149] Depending on the embodiment, a predetermined distance in the first direction D**1** may be 5 mm, and a predetermined width in the second direction D**2** may be 10 mm.
[0150] FIG. **9** and FIG. **10** are diagrams that omit the head **110** disposed between the conditioner arm **140** and the disk **130** for the purpose of illustration.
[0151] When the disk **130** is detected throughout the entire width in the second direction D**2** as shown in FIG. **9**, a step S**210** in which the judgment part **700** determines that the conditioner arm **140** is in correct position occurs.
[0152] When the judgment part **700** determines that the conditioner arm **140** is in correct position, a step S**220** in which the control part **900** adjusts the polishing pad conditioner **100** to condition the polishing pad **1** may be included.
[0153] As described above, in order for the polishing pad conditioner **100** to condition the polishing pad **1**, it is first necessary to move the position of the polishing pad conditioner **100** from the cup position to a position in which the disk **130** contacts the upper surface of the polishing pad **1**, for example, by the rotation of the driving shaft **150**.
[0154] FIG. **10** shows a case where the conditioner arm **140** is not placed in correct position (step S**230**), and the polishing pad conditioner position detection module **10** cannot detect the disk **130** within a predetermined distance in the first direction D**1** and/or within a predetermined width in the

second direction D**2** perpendicular to the first direction D**1**. FIG. **10** in particular shows a case where the disk **130** is not detected at all within a predetermined width in the second direction D**2**.

[0155] As shown in FIG. **10**, the disk **130** may not be detected at all within a predetermined width in the second direction D**2**, but on the contrary, step S**230** may occur in a case in which the disk **130** is detected only in an area that is only partially located within a predetermined width in the second direction D**2**.

[0156] As described above, when the disk **130** is detected only in a partial area, the detection part **400** does not detect the disk **130** as properly covering the width in the second direction D**2**. For example, in some embodiments, the disk **130** may only be detected in a partial area is because the conditioner arm **140** is tilted from its correct position.

[0157] When the disk **130** is not detected within a predetermined width in the second direction D**2** as shown in FIG. **10**, in step S**230**, the judgment part **700** determines that the position of the conditioner arm **140** is incorrect or misaligned.

[0158] In one embodiment, when the disk **130** is not detected within a predetermined width in the second direction D**2**, step S**240** occurs, in which the alarm part **800** generates an alarm.

[0159] As described above, there should be no interference between the polishing pad conditioner position detection module **10** and other previously used CMP components in how the polishing pad conditioner position detection module **10** detects the position of the polishing pad conditioner **100**.

[0160] Specifically, the polishing pad conditioner position detection module **10** should have no interference with the platen shield surrounding the platen **22** and there should be no interference with the injection nozzle **210** of the cleaning module **200** and the injection part (not shown) described above. In addition, there should be no collision with the polishing pad conditioner position detection module **10** while the polishing pad conditioner **100** rotates by the driving shaft **150**.

[0161] In the conditioning process of the conventional polishing pad conditioner, the disk may be disengaged, or the disk may be placed in a position different from the target position on the CMP polishing pad due to problems such as deterioration of parts of the conditioner arm. As a result, there was a problem of conditioning the CMP polishing pad with the conditioner disk positioned off-center on the CMP polishing pad. In those cases, it would often be necessary to rely on human technicians to detect misalignment of the conditioner disk, and such detection might not occur until significant damage has been done to polishing pads and/or devices being manufactured.

[0162] The polishing pad conditioner position detection module **10** according to the present disclosure detects in advance whether the detection part **400** is in a state where the disk **130** is mounted on the conditioner **100** and whether there is a problem with the movement of the conditioner arm **140**, thereby preventing the CMP polishing pad **1** from being worn unevenly and improving the quality of the CMP polishing pad **1**.

[0163] In particular, the polishing pad conditioner position detection module **10** according to the present disclosure is an additional configuration that is combined with the cleaning module **200** for cleaning the polishing pad conditioner **100**. Therefore, the polishing pad conditioner position detection module **10** according to the present disclosure may be highly useful in that it can be utilized while using the existing polishing pad conditioner **100** without any modification.

[0164] Although the embodiments of the present disclosure have been described above, the embodiments are not limited thereto, and it is possible to implement various modifications within the scope of the claims, detailed explanations, and accompanying drawings.

Claims

1. A polishing pad conditioner position detection module for a polishing device which includes a polishing pad conditioner, a disk for conditioning a polishing surface on a polishing pad, and a cleaning module for cleaning the disk and placed adjacent to the polishing pad, the polishing pad conditioner position detection module comprising: a frame configured to surround the cleaning module; a housing connected to the frame; and a detector disposed inside the housing and configured to face a side of the disk when the disk is attached to the polishing pad conditioner, wherein the detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in a correct predetermined position.

2. The polishing pad conditioner position detection module of claim 1, wherein: the detector is configured to detect whether the disk is mounted on the polishing pad conditioner and when mounted, whether the disk is in the correct predetermined position while a portion of polishing pad conditioner configured to hold the disk is positioned in a standby position adjacent to the cleaning module.

3. The polishing pad conditioner position detection module of claim 1, wherein: the detector is positioned to detect whether the disk is in the correct predetermined position based on the disk being mounted on a holder located underside a head of the polishing pad conditioner.

4. The polishing pad conditioner position detection module of claim 1, wherein: one end of the detector is arranged to face a first direction toward a center of the disk when the disk is in the correct predetermined position and is configured to detect the disk positioned within a predetermined distance in the first direction.

5. The polishing pad conditioner position detection module of claim 4, wherein: the detector is configured to detect whether the disk is located within a predetermined location having a width in a second direction which is perpendicular to the first direction.

6. The polishing pad conditioner position detection module of claim 1, further comprising a cable connected to the detector and extending out of the housing.

7. The polishing pad conditioner position detection module of claim 6, wherein the cable comprises: a first cable portion connected to the detector and placed inside the housing; and a second cable portion connected to the first cable portion and extending out of the housing.

8. The polishing pad conditioner position detection module of claim 6, further comprising a receiving part connected to the housing for receiving the cable.

9. The polishing pad conditioner position detection module of claim 8, wherein the receiving part further comprises a hole for routing the cable externally.

10. The polishing pad conditioner position detection module of claim 1, wherein the housing is connected to outer surface of the cleaning module.

11. The polishing pad conditioner position detection module of claim 1, wherein the housing further comprises a guide part inside which the detector is placed.

12. The polishing pad conditioner position detection module of claim 1, wherein the detector and the housing in which the detector is placed are positioned to protrude from the cleaning module toward the disk.

13. A polishing device comprising: a rotatable platen; a polishing pad placed on an upper surface of the platen and having a polishing surface; a polishing head disposed above the polishing pad and configured for mounting a substrate on a bottom surface of the polishing head; a driver configured to rotate the platen with respect to an axis perpendicular to the polishing surface of the polishing pad; a slurry supply outlet configured to supply slurry to the polishing surface of the polishing pad; a polishing pad conditioner for conditioning the polishing surface of the polishing pad; and a polishing pad conditioner position detection module configured to detect whether a disk of the polishing pad conditioner is correctly aligned.

14. A polishing pad conditioner position detection method for detecting whether a polishing pad conditioner including a conditioner arm, a head connected to the conditioner arm, and a disk mounted on the head is in correct position, the method comprising: placing the polishing pad conditioner in cup position adjacent to a cleaning module for cleaning the disk; and detecting, using a detector placed on the cleaning module, whether a center of the disk is located within a predetermined distance in a first direction from the detector.

15. The polishing pad conditioner position detection method of claim 14, comprising: detecting whether the disk is located within a predetermined distance in the first direction and within a predetermined width in a second direction perpendicular to the first direction.

16. The polishing pad conditioner position detection method of claim 15, comprising: determining that the disk is disposed on a conditioner arm when the disk is detected within a predetermined distance in the first direction; and determining that the conditioner arm is in the correct position when the disk is detected within a predetermined width in the second direction.

17. The polishing pad conditioner position detection method of claim 16, comprising: determining that the disk is separated from a holder placed underside a head of the polishing pad conditioner when the disk is not detected within a predetermined distance in the first direction.

18. The polishing pad conditioner position detection method of claim 15, comprising: determining that a position of the conditioner arm is displaced when the disk is not detected within a predetermined width in the second direction.

19. The polishing pad conditioner position detection method of claim 14, comprising: generating an alarm when the disk is not detected within a predetermined distance in the first direction.